2022

# Lecture 3 & 4: SVM (Part I) and KKT Conditions

Week 3 & Week 4

*Lecturer: Tianyu Wang*

## 1 Maximizing the Margins

To start with, we consider a linearly separable dataset $\{(x_i, y_i)\}_{i=1}^n$. Linear separability is equivalent to existence of a hyperplane that perfectly classifies the dataset. Consider a linear model $f(x) = \begin{cases} +1, & \text{if } w^\top x + b \geq 0, \\ -1, & \text{otherwise.} \end{cases}$ This function defines a hyperplane $h(x) = w^\top x + b$. We want all observations to be far away from the decision boundary. In machine learning, margins are unsigned distances from the data points to the decision boundary.[1]

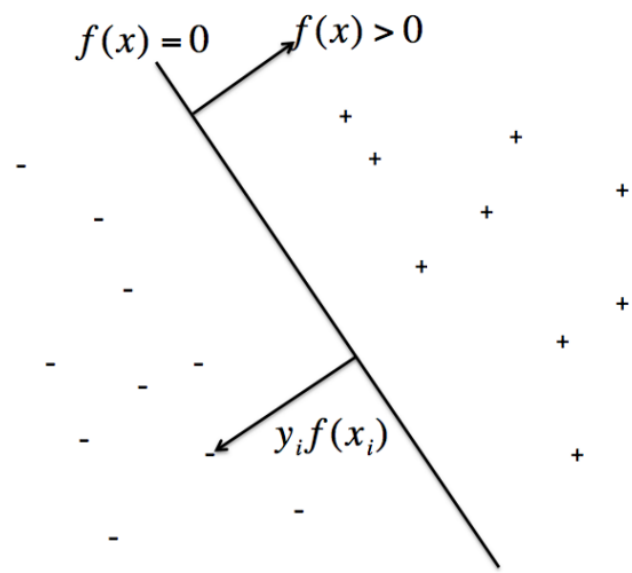


Figure 1: Margins. Picture Credit: C. Rudin.

Let $\gamma$ be the signed distance from $x \in \mathbb{R}^d$ to the hyperplane $w^\top x + b = 0$. Then we know

[1] Some authors uses different definitions for "margin", but it's always the case that the margin summarizes distances from the data points to the decision boundary.

$x - \gamma \frac{w}{\|w\|_2}$ is on the plane $w^\top x + b = 0$. Thus we have

$$w^\top \left( x - \gamma \frac{w}{\|w\|_2} \right) + b = 0,$$

which gives

$$\gamma = \frac{w^\top x + b}{\|w\|_2}.$$

On a (linearly separable) dataset $\{(x_i, y_i)\}_{i=1}^n$. The objective for margin maximization is

$$\max_{w,b,\gamma} \gamma, \quad \text{such that} \quad y_i \frac{w^\top x_i + b}{\|w\|_2} \geq \gamma \quad , \forall i = 1, 2, \cdots, n.$$

Note that the constraint satisfaction is invariant to scaling. Thus we can set $\|w\|_2 = \frac{1}{\gamma}$ and the above optimization problem becomes

$$\max_{w,b} \frac{1}{\|w\|_2}, \quad \text{such that} \quad y_i \left( w^\top x_i + b \right) \geq 1 \quad , \forall i = 1, 2, \cdots, n.$$

This problem can be further relaxed to

$$\min_{w,b} \frac{1}{2} \|w\|_2^2, \quad \text{such that} \quad y_i \left( w^\top x_i + b \right) \geq 1 \quad , \forall i = 1, 2, \cdots, n. \tag{1}$$

To solve this problem, we need some knowledge on constrained convex optimization.

# 2 Basics of Constrained Convex Optimization and KKT Conditions

Consider a convex optimization problem

$$\begin{aligned} & \min_{x \in \mathbb{R}^n} f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \ i = 1, 2, \cdots, m \\ & h_i(x) = 0, i = 1, \cdots, p, \end{aligned}$$

where $f : \mathbb{R}^n \to \mathbb{R}$, $g_i : \mathbb{R}^n \to \mathbb{R}$ are differentiable convex functions, and $h_i : \mathbb{R}^n \to \mathbb{R}$ are linear functions.

Alternatively, one can replace the equality constraints, since $h_i(x) = 0$ *iff* $h_i(x) \geq 0$ and $h_i(x) \leq 0$. For simplicity, we omit the equality constraints.

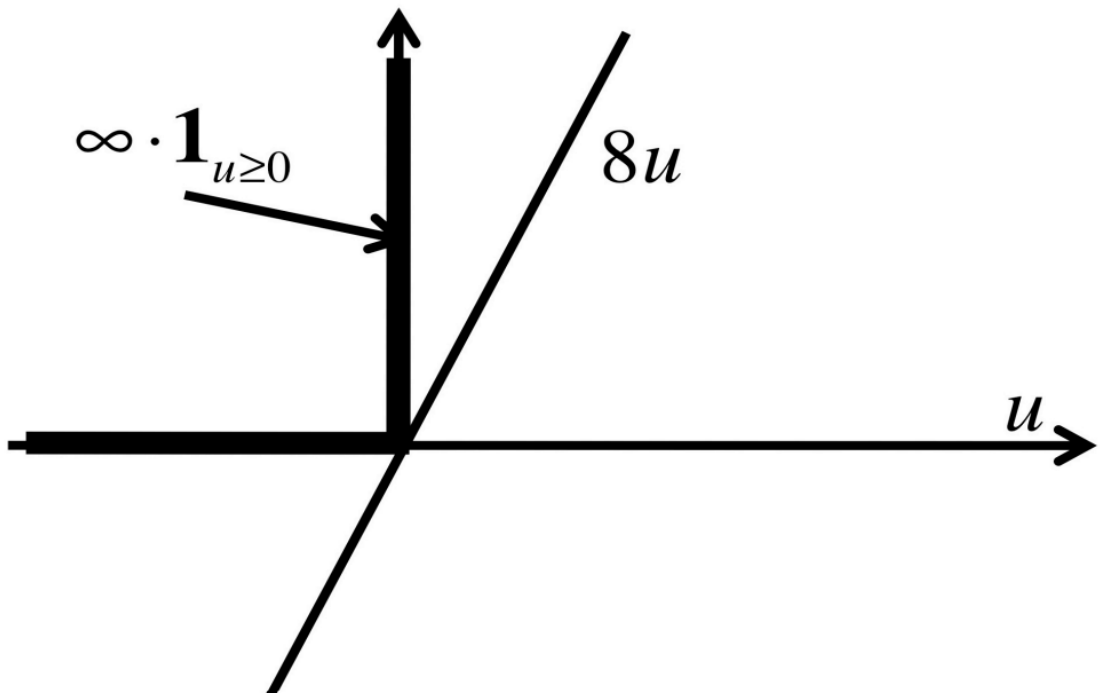


Figure 2: Approximation of the $\infty \cdot \mathbb{I}_{[x>0]}$ function. Picture credit: C. Rudin.

## 2.1 Intuition behind the Lagrangian

If we receive an infinitely large penalty whenever a constraint is violated, the objective for the above optimization problem can be written as

$$f(x) + \infty \cdot \sum_{i=1}^{m} \mathbb{I}_{[g_i(x)>0]}. \tag{2}$$

The $\infty \cdot \mathbb{I}_{[x>0]}$ function can be approximated by a linear function, as shown in Figure 2.

We can approximate $\infty \cdot \mathbb{I}_{[x>0]}$ by $\alpha x$ ($\alpha \geq 0$). By using this approximation, we get the Lagrangian for the optimization problem:

$$\boldsymbol{L}(x, \alpha) = f(x) + \sum_{i=1}^{m} \alpha_i g_i(x),$$

where $\alpha$ are called dual variables.

The primal objective can be written as

$$\Theta_p(x) := \max_{\alpha_i \geq 0, i=1,2,\cdots,m} \boldsymbol{L}(x, \alpha),$$

and the dual objective can be written as

$$\Theta_d(\alpha) := \min_x \boldsymbol{L}(x, \alpha)$$

## 2.2 Duality Properties

**Proposition 2.1** (max-min inequality)**.** *Consider a real-valued function $\phi$ and two sets $X, Y$ on which $\phi$ is defined. It holds that*

$$\max_{y \in Y} \min_{x \in X} \phi(x, y) \leq \min_{x \in X} \max_{y \in Y} \phi(x, y).$$

*Proof.* For any fixed $x, y$, it holds that

$$\min_{x' \in X} \phi(x', y) \leq \max_{y' \in Y} \phi(x, y').$$

Since the above inequality holds for any $(x, y)$, we finish the proof by taking the minimum over $x \in X$on the right-hand side, then the maximum over $y \in Y$ on the left-hand side. □

By the max-min inequality, we know that

$$\max_{\alpha \geq 0, i=1,2,\cdots,m} \Theta_d(\alpha) = \max_{\alpha \geq 0, i=1,2,\cdots,m} \min_x \boldsymbol{L}(x, \alpha) \leq \min_x \max_{\alpha \geq 0, i=1,2,\cdots,m} \boldsymbol{L}(x, \alpha) = \min_x \Theta_p(x),$$

which is known as weak duality.

Let $p^*$ be the optimal value of the primal problem and let $d^*$ be the optimal value of the dual problem. By weak duality, the number $p^* - d^*$ is called *duality gap*.

When the equality holds, we know strong duality. That is, the optimal value of the dual objective equals the optimal value of the primal objective. Next we discuss strong duality. A sufficient condition for strong duality is called Slater's condition.

**Theorem 2.2** (Strong duality)**.** *When the constraint sets has one strict interior point, then strong duality holds. More formally, if there exists $\widetilde{x}$ such that $g_i(\widetilde{x}) < 0$ for all inequality constraints $i = 1, 2, \cdots, m$ and $h_j(\widetilde{x}) = 0$ for all equality constraints $j = 1, 2, \cdots, p$ (Slater's condition), then*

$$\max_{\alpha \geq 0, i=1,2,\cdots,m} \Theta_d(\alpha) = \min_x \Theta_p(x).$$

The proof idea is illustrated in Figure 3. We will provide a proof for the the case where there is no equality constraints. The proof for cases with equality constraints uses a similar argument.

*Proof.* Consider the set

$$V = \{(u, w) \in \mathbb{R}^m \times \mathbb{R} : \exists x, \text{ such that } f(x) \leq w \text{ and } g_i(x) \leq u_i \ \forall i = 1, 2, \cdots, m\}.$$

**Step 1.** By convexity of $f$ and $g_i$, the set $V$ is convex. This can be verified by definition.

Figure 3: Strong duality proof illustration.

**Step 2.** Let $p^*$ be the solution to the primal problem. The point $(0, p^*)$ is on the boundary of $V$. Otherwise there is a contradiction to the optimality of $p^*$.

By Claim 1 and Claim 2, there exist $(\mu, \mu_0) \in \mathbb{R}^m \times \mathbb{R}$ and $(\mu, \mu_0) \neq 0$ such that

$$\mu^\top u + \mu_0 w \geq 0^\top u + \mu_0 p^* = \mu_0 p^* \tag{3}$$

for all $(u, w) \in V$.

**Step 3.** It holds that $\mu_i \geq 0$ for all $i = 0, 1, 2, \cdots, m$. Note that (3) implies

$$\mu^\top u + \mu_0 (w - p^*) \geq 0, \quad \forall (u, w) \in V.$$

Since $u_i, w$ can be arbitrarily large, $\mu_i < 0$ would contradict to the above inequality.

**Step 4.** It holds that $\mu_0 \neq 0$.

Suppose, in order to get a contradiction, that $\mu_0 = 0$. In this case

$$\inf_{(u,w) \in V} \mu^\top u \geq 0.$$

At the same time,

$$\inf_{(u,w) \in V} \mu^\top u = \inf_x \sum_{i=1}^m \mu_i g_i(x) \leq \sum_{i=1}^m \mu_i g_i(\widetilde{x}) < 0,$$

where the last inequality uses Claim 3 and the Slater's condition.

**Step 5.** Finish up the proof.

Let $\alpha = \frac{\mu}{\mu_0}$. Note that $\alpha_i \geq 0$ for all $i = 1, 2, \cdot, m$. Plugging this back to (3) implies that

$$\alpha^\top u + w \geq p^*,$$

for all $(u, w) \in V$. Thus we have

$$\Theta_d(\alpha) = \inf_x \left( f(x) + \sum_{i=1}^m \alpha_i g_i(x) \right) = \inf_{u,w} \left( \alpha^\top u + w \right) \geq p^*, \tag{4}$$

which implies

$$\max_{\alpha_i \geq 0, i=1,2,\cdots,m} \Theta_d(\alpha) \geq p^* = \min_x \Theta_p(x). \tag{5}$$

Together with the weak duality theorem, the above result finishes the proof.

□

For cases with equality constraints, one can create another set of variables for the equality constraints, and the general idea is similar.

## 2.3 KKT Conditions

**Theorem 2.3** (Karush-Kuhn-Tucker)**.** *Consider a constraint convex optimization problem where strong duality holds. The following conditions are satisfied at* $(x^*, \alpha^*)$

- *Primal feasibility:* $g_i(x^*) \leq 0$ *for all* $i = 1, 2, \cdots, m$*;*
- *Dual feasibility:* $\alpha_i^* \geq 0$ *for all* $i = 1, 2, \cdots, m$*;*
- *Stationarity:* $\nabla_x \boldsymbol{L}(x^*, \alpha^*) = 0$*;*
- *Complementary Slackness:* $\alpha_i^* g_i(x^*) = 0$ *for all* $i = 1, 2, \cdots, m$*,*

*if and only if* $x^*$ *optimally solves the primal problem and* $\alpha^*$ *optimally solves the dual problem.*

The four conditions (primal/dual feasibility, stationarity, complementary slackness) are called Karush-Kuhn-Tucker (KKT) conditions.

*Proof.* **Necessity.** Let $x^*$ and $\alpha^*$ be primal and dual solutions with zero duality gap, then

$$f(x^*) = \Theta_p(x^*) = \boldsymbol{L}(x^*, \alpha^*) \overset{(i)}{=} \min_x \{ f(x) + \sum_{i=1}^m \alpha_i^* g_i(x) \} \leq f(x^*) + \sum_{i=1}^m \alpha_i^* g_i(x^*) \overset{(ii)}{\leq} f(x^*).$$

Thus all of the above inequalities are actually equalities. Thus we have

- Primal feasibility and dual feasibility, which directly follows from that $x^*$ and $\alpha^*$ are primal solution and dual solution.
- Stationarity, since $x$ minimizes $\boldsymbol{L}(x, \alpha^*)$ (by Eq. $(i)$).
- Complementary Slackness, by Eq. $(ii)$.

**Sufficiency.** Let the KKT conditions hold. We have

$$\min_x \boldsymbol{L}(x, \alpha^*) \overset{(iii)}{=} \boldsymbol{L}(x^*, \alpha^*) = f(x^*) + \sum_{i=1}^m \alpha_i^* g_i(x^*) \overset{(iv)}{=} f(x^*),$$

where $(iii)$ uses stationarity and $(iv)$ uses complementary slackness. □

# 3 Back to SVM

Now we have the tools we need to solve for the SVM objective. Recall the SVM objective is

$$\min_{w,b} \frac{1}{2}\|w\|_2^2, \quad \text{such that} \quad y_i\left(w^\top x_i + b\right) \geq 1 \quad , \forall i = 1, 2, \cdots, n.$$

The Lagrangian is

$$\boldsymbol{L}(w, b, \alpha) = \frac{1}{2}\|w\|_2^2 + \sum_{i=1}^{n} \alpha_i (1 - y_i\left(w^\top x_i + b\right)).$$

The KKT conditions are

- $1 - y_i(w^\top x_i + b) \leq 0$ for $i = 1, 2, \cdots, n$. (Primal feasibility)
- $\alpha_i \geq 0$ for $i = 1, 2, \cdots, n$. (Dual feasibility)
- $w - \sum_{i=1}^{n} \alpha_i y_i x_i = 0$ and $\sum_{i=1}^{n} \alpha_i y_i = 0$. (Stationarity)
- $\alpha_i(1 - y_i\left(w^\top x_i + b\right))$ for $i = 1, 2, \cdots, n$. (Complementary slackness)

The KKT conditions are satisfied at the optimal solution $(w^*, b^*, \alpha^*)$.

## 3.1 Support Vectors

Look at the complementary slackness condition (and feasibility conditions):

$$\alpha_i(1 - y_i\left(w^\top x_i + b\right)) = 0 \Rightarrow \begin{cases} 1 - y_i\left(w^\top x_i + b\right) < 0 \ \ \& \ \ \alpha_i = 0 & \text{(inactive constraint)} \\ 1 - y_i\left(w^\top x_i + b\right) = 0 \ \ \& \ \ \alpha_i > 0 & \text{(active constraint)} \end{cases}.$$

The support vectors are those data points whose constraint is active (See Figure 4).

## 3.2 Solve for the SVM model

As shown before, the optimal SVM model is specified by $(w^*, b^*)$ that solves the constrained optimization problem. The optimal solution $(w^*, b^*)$ satisfies

$$w^* = \sum_{i=1}^{n} \alpha_i^* y_i x_i, \quad b^* = y_{i_0} - x_{i_0}^\top w^* \text{ for some support vector } x_{i_0}.$$

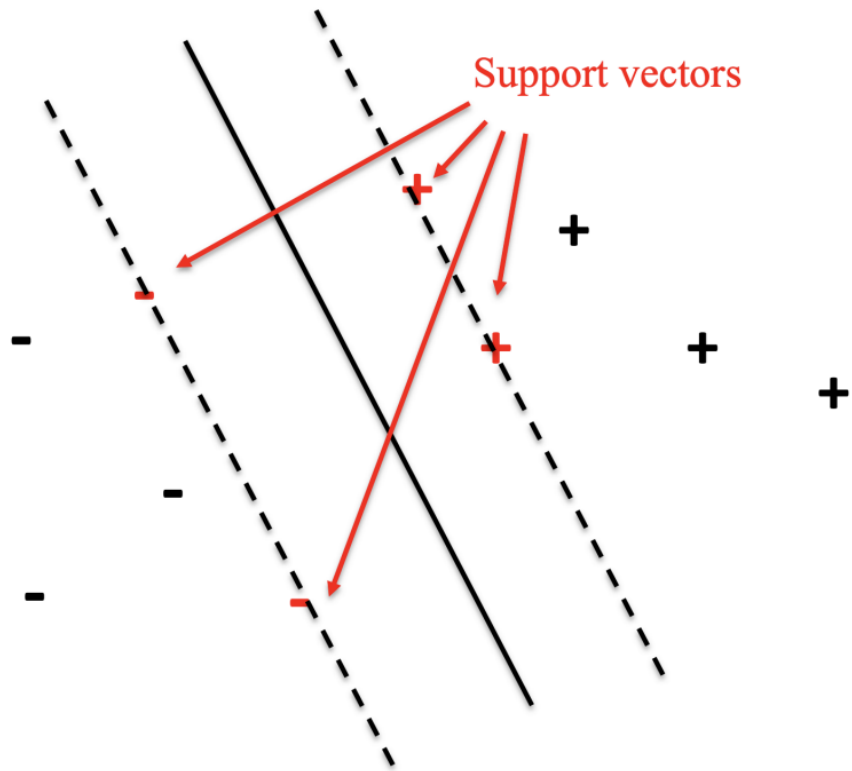


Figure 4: Support Vectors. Picture Credit: C. Rudin.

Thus the dual solution $\alpha_i^*$ determines the model. The Lagrangian is

$$
\begin{aligned}
\boldsymbol{L}(w^*, b^*, \alpha^*) &= \frac{1}{2}\|w^*\|_2^2 + \sum_{i=1}^n \alpha_i^* \left(1 - y_i \left(x_i^\top w^* + b\right)\right) \\
&= \frac{1}{2}\|w^*\|_2^2 + \sum_{i=1}^n \alpha_i^* - \sum_{i=1}^n \alpha_i^* y_i x_i^\top w^* + \sum_{i=1}^n \alpha_i^* y_i b^* \\
&= \frac{1}{2}\|w^*\|_2^2 + \sum_{i=1}^n \alpha_i^* - \sum_{i=1}^n \alpha_i^* y_i x_i^\top w^* && \left(\textstyle\sum_{i=1}^n \alpha_i^* y_i = 0\right) \\
&= \frac{1}{2}\|w^*\|_2^2 + \sum_{i=1}^n \alpha_i^* - \|w^*\|_2^2 && \left(w^* = \textstyle\sum_{i=1}^n \alpha_i^* y_i x_i\right) \\
&= \sum_{i=1}^n \alpha_i^* - \frac{1}{2}\|w^*\|_2^2 \\
&= \sum_{i=1}^n \alpha_i^* - \sum_{i=1}^n \sum_{j=1}^n \alpha_i^* \alpha_j^* y_i y_j x_i^\top x_j^\top .
\end{aligned}
$$

Thus the dual objective is

$$\Theta_d(\alpha) = \sum_{i=1}^{n} \alpha_i - \sum_{i=1}^{n}\sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j x_i^\top x_j^\top \quad \text{subject to} \begin{cases} \alpha_i \geq 0,\ i = 1, 2, \cdots, n \\ \sum_{i=1}^{n} \alpha_i y_i = 0. \end{cases} \tag{6}$$

The dual problem is often easier to solve in practice, since the constraints are simpler than the primal problem.

Next time we will continue on SVM.


## Acknowledgement

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